Astroshop capacity review — Q3 2026

Scope

This review covers the tier-0 request path only: cart, checkout, frontend, frontend-proxy, payment, product-catalog. Tier-1 and tier-2 services are reviewed on a separate cadence.

Method

Peak concurrent virtual users over the quarter were replayed against a staging deployment at 1x, 2x and 4x. Headroom is defined as the multiple of current peak at which p99 latency first exceeds the service SLO.

Frontend

The frontend fans out to every product-path service, so its own capacity is bounded by its slowest dependency under load, not by its own CPU. At 2x it stayed within SLO; at 4x, p99 was dominated by product-catalog and recommendation queueing.

Checkout

Checkout is synchronous over six dependencies, so its latency is the sum of theirs plus its own overhead. Headroom is 2.5x. Payment and shipping are the two that degrade first under load; email does not matter because the call is best-effort.

Product-catalog

Shared by frontend and recommendation. At 4x frontend load plus the recommendation-cache-failure fan-out pattern, product-catalog is the first tier-0 service to breach SLO. This is the single highest-risk capacity finding.

Recommendation

The in-process cache is load-bearing for product-catalog's capacity, not just recommendation's latency. A recommendation deploy that resets the cache briefly doubles product-catalog read load — schedule deploys off-peak.

Kafka and consumers

accounting and fraud-detection lag is a function of order rate. At 4x order rate, steady-state lag stayed under the 60s freshness SLO but recovery from a 5-minute broker stall took 12 minutes. Consider a second partition.

Recommendations

1. Add a product-catalog read replica before Q4 peak. 2. Move recommendation deploys to a maintenance window. 3. Add a second orders-topic partition. 4. Re-run this review at 6x to find the next bottleneck.